

INTERNSHIP REPORT

A dissertation submitted in the partial fulfillment of the
requirement for the Industry Internship Program

Submitted

By

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PG DEPARTMENT OF COMPUTER SCIENCE
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CERTIFICATE

This is to certify that the internship was done at the company

“Cadd Tecnolynx”

by

VIKASHINI S T

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As the partial fulfillment of the requirement for Industry Internship Program in **MASTER OF COMPUTER SCIENCE** during the academic year 2025-2027.

Submitted for the University Examination held on _____.

Internal Guide

Head of the Department

Place: Thoothukudi

External Examiner

ACKNOWLEDGEMENT

It is with great humility that I thank God Almighty for guiding me throughout my internship at Cadd Technolynx. I am deeply grateful to my friends and family, whose love and encouragement have given me renewed purpose and motivation to complete this internship.

It is the right time for me to express my heartiest thankfulness to our esteemed Institution, **A.P.C. Mahalaxmi College for Women, Thoothukudi**, for providing me with technical knowledge and all facilities in the field of Computer Science.

I am privileged to owe my sincere thanks to my beloved Principal **Dr. K. Subbulakshmi, M.A., M.Phil., Ph.D.**, who always cares to fulfill the needs of the students and be a moral support to me.

I whole heartedly thank my Head of the Department **Dr. V. Shyamala Susan, M.C.A., M.Phil., Ph.D.**, for providing me with constructive criticism and suggestions to fulfill my internship.

I extend my gratitude to guide **Mrs. J. AROCKIA JEYANTHI, MCA, SLET., NET.**, for her valuable guidance and encouragement throughout my internship.

It is my profound duty to thank all my department faculties who have helped me in various activities.

I extend my gratitude to Cadd Tecnolynx Technical Head **Mr. Vinoth Kumar B**, for his valuable guidance and encouragement throughout my internship.

ABSTRACT

CADD Technolynx is a professional training and skill development organization that provides industry-oriented training programs in various domains, including Data Analytics, Software Development, Full-Stack Development, CADD Technologies, and emerging IT fields. The institute focuses on delivering practical knowledge through hands-on projects, real-world datasets, and industry-relevant tools to enhance students' technical and analytical skills.

The primary focus of my internship was to gain practical exposure to Data Analytics concepts and tools through project-based learning. During the internship, I learned Python programming fundamentals and developed a Shopping Mart Menu and Cart Management application as part of my programming practice. I was also assigned a Python-based project, which was successfully completed and hosted as a web application.

Further, I explored the fundamentals of Data Analytics, including the data analytics lifecycle and analytics pipeline. Using Jupyter Notebook, I performed data exploration and visualization activities to derive meaningful insights from datasets. The internship also introduced me to Tableau, where I learned to import datasets from Kaggle, create various visualizations, and develop interactive dashboards for effective data presentation.

As a capstone activity, I designed a dynamic Tableau dashboard and presented the project findings through an online presentation supported by a PowerPoint presentation. The internship concluded with a valedictory session and doubt-clearing discussion, providing valuable feedback and guidance.

This internship helped me develop practical skills in Python programming, data analysis, data visualization, dashboard creation, and project presentation. It also provided valuable exposure to the complete data analytics workflow, strengthening my understanding of business intelligence tools and reinforcing my interest in pursuing a career in Data Analytics and Data Science.

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LEARNING OBJECTIVES / INTERNSHIP OBJECTIVES

- To gain comprehensive knowledge of Data Analytics concepts, methodologies, and industry practices by understanding the complete analytics lifecycle, including data collection, data preprocessing, data analysis, visualization, and interpretation of results for effective decision-making.
- To develop practical programming skills using Python and Jupyter Notebook by implementing real-world applications, completing project assignments, and learning how programming can be utilized to solve analytical and business-related problems.
- To acquire hands-on experience in data visualization and business intelligence tools such as Tableau by working with real datasets, creating meaningful visualizations, designing interactive dashboards, and presenting analytical insights effectively.
- To enhance analytical thinking, problem-solving abilities, and technical competency through project-based learning, enabling the identification of patterns, trends, and actionable insights from structured datasets.
- To strengthen professional communication, presentation, and teamwork skills by preparing project reports, delivering presentations, participating in discussions, and gaining exposure to industry-oriented workflows and best practices in the field of Data Analytics.

OVERVIEW OF INTERNSHIP ACTIVITIES

S.No.	DATE	DAY	NAME OF THE TOPIC
1	11.04.2026	Saturday	Internship Orientation and Introduction Session
2	13.04.2026	Monday	Python Installation and Environment Setup
3	16.04.2026	Thursday	Python Basics
4	17.04.2026	Friday	Control Statements and Loops in Python
5	18.04.2026	Saturday	Python Object-Oriented Concepts
6	19.04.2026	Sunday	Development of Shopping Mart Program
7	20.04.2026	Monday	Presentation of Shopping Mart Application
8	21.04.2026	Tuesday	Introduction to Data Analytics
9	22.04.2026	Wednesday	Understanding the Data Analytics Lifecycle
10	23.04.2026	Thursday	Data Analytics Pipeline
11	24.04.2026	Friday	Python Project – Requirement Analysis
12	25.04.2026	Saturday	Python Project Development
13	27.04.2026	Monday	Python Project Presentation and Testing
14	28.04.2026	Tuesday	Project Completion and Documentation
15	29.04.2026	Wednesday	Project Deployment and Hosting
16	30.04.2026	Thursday	Introduction to Jupyter Notebook Environment
17	02.05.2026	Saturday	Data Analysis Using Jupyter Notebook
18	04.05.2026	Monday	Data Cleaning and Preprocessing Concepts
19	05.05.2026	Tuesday	Exploratory Data Analysis (EDA) Techniques
20	19.05.2026	Tuesday	Data Visualization Fundamentals
21	20.05.2026	Wednesday	Creating Visualizations Using Python Libraries
22	21.05.2026	Thursday	Introduction to Tableau and Installation
23	22.05.2026	Friday	Understanding Tableau Interface and Features
24	23.05.2026	Saturday	Importing Kaggle Dataset into Tableau
25	25.05.2026	Monday	Creating Basic Charts and Visualizations
26	26.05.2026	Tuesday	Advanced Visualizations and Data Exploration
27	27.05.2026	Wednesday	Building Interactive and Dynamic Dashboards
28	29.05.2026	Friday	Dashboard Refinement
29	30.05.2026	Saturday	Online Project Presentation
30	31.05.2026	Sunday	Valedictory Session and Doubt Clarification

INTRODUCTION

This report outlines my internship experience at CADD Technolynx, where I worked as a Data Analytics Intern. During this internship, I gained practical exposure to Python programming, data analytics methodologies, data visualization techniques, and business intelligence tools. The internship focused on developing analytical thinking and technical skills through hands-on projects, real-world datasets, and dashboard development.

In today's digital world, organizations generate vast amounts of data from various sources. However, raw data alone has little value unless it is processed, analyzed, and transformed into meaningful information. Data Analytics plays a crucial role in helping businesses identify trends, improve decision-making, optimize operations, and gain a competitive advantage.

The internship provided a comprehensive understanding of the Data Analytics lifecycle, including data collection, data cleaning, data processing, analysis, and visualization. Initially, I completed a Python project on “**Stock Profit AI Optimizer**” and successfully hosted the application, gaining practical experience in project deployment.

Further, I explored data analytics concepts using Jupyter Notebook, where I performed data exploration and visualization tasks. To enhance my business intelligence skills, I worked with Tableau by importing datasets from Kaggle, creating various charts and visualizations, and developing an “**Interactive Dashboard**”. The project concluded with an online presentation of the dashboard and findings using PowerPoint slides.

The internship provided valuable hands-on experience with Python, Jupyter Notebook, Tableau, and data visualization techniques. It enhanced my analytical, technical, and presentation skills while giving me practical insight into the complete data analytics workflow. Overall, this experience strengthened my understanding of data-driven decision-making and reinforced my interest in pursuing a career in Data Analytics, Business Intelligence, and Data Science.

SYSTEM ANALYSIS

Traditionally, data analysis, stock market evaluation, and business reporting were performed using manual methods and spreadsheet-based tools. These approaches required significant effort for data collection, cleaning, processing, and report generation. Analyzing stock performance manually was time-consuming and often led to inaccurate investment decisions, while traditional reporting methods made it difficult to identify trends and gain meaningful insights from large datasets. Static reports and spreadsheets also lacked interactive features, limiting the effectiveness of data-driven decision-making.

With the advancement of data analytics and business intelligence technologies, modern tools such as Python, Jupyter Notebook, and Tableau provide efficient solutions for data processing, analysis, prediction, and visualization. The Stock Profit AI Optimizer utilizes Python to analyze stock data and identify potential profit opportunities, while the Data Analytics Pipeline implemented in Jupyter Notebook enables systematic data cleaning, processing, and exploratory analysis. Tableau further enhances the analytical process by transforming datasets into interactive dashboards and visual reports, allowing users to explore trends and insights effectively. Together, these technologies improve accuracy, reduce manual effort, and support informed decision-making through data-driven solutions.

EXISTING SYSTEM

Traditionally, stock analysis, data processing, and business reporting were performed using manual methods and spreadsheet applications. Analysts and investors had to collect, organize, and evaluate large amounts of data manually, making the process time-consuming and prone to errors. Generating reports and identifying trends required significant effort, while static spreadsheets provided

limited visualization capabilities. These limitations often resulted in delayed decision-making and reduced efficiency in extracting meaningful insights from data.

PROPOSED SYSTEM

The proposed system integrates Python, Jupyter Notebook, and Tableau to create a comprehensive data analytics solution. The Stock Profit AI Optimizer utilizes Python-based analytical techniques to evaluate stock performance and identify potential profit opportunities. Jupyter Notebook is used to implement a structured data analytics pipeline involving data collection, cleaning, processing, analysis, and visualization. Tableau enhances the analytical process by transforming datasets into interactive charts and dynamic dashboards, enabling users to explore insights effectively. This integrated approach improves accuracy, reduces manual effort, and supports faster, data-driven decision-making through advanced analytics and visualization techniques.

SOFTWARE REQUIREMENTS & SPECIFICATIONS

HARDWARE SPECIFICATION

Stock Profit AI Optimizer (Python) Processor:

Processor: Intel Core i3 or equivalent.

RAM: Minimum 4 GB (8 GB recommended).

Storage: 1 GB free disk space.

Display: Standard monitor with 1366×768 resolution or higher.

Internet Connectivity: Stable broadband or Wi-Fi connection for accessing stock datasets and online resources.

Data Analytics Pipeline (Jupyter Notebook):

Processor: Intel Core i5 or equivalent.

RAM: Minimum 8 GB.

Storage: 2 GB free disk space for datasets and notebook files.

Display: Standard monitor with 1366×768 resolution or higher.

Internet Connectivity: Required for downloading datasets, libraries, and project resources.

Tableau Dashboard Development:

Processor: Intel Core i5 or equivalent.

RAM: Minimum 8 GB (16 GB recommended for large datasets).

Storage: 2 GB free disk space for Tableau installation and dashboard files.

Graphics: Standard graphics support for data visualization.

Display: 1366×768 resolution or higher.

Internet Connectivity: Required for dataset access, dashboard publishing, and online presentations.

SOFTWARE SPECIFICATION

Operating System: Windows 10/11, Linux, or macOS.

Programming Language: Python 3.x.

Development Environment: Jupyter Notebook.

Libraries Used: Pandas, NumPy, Matplotlib, Seaborn, and other data analytics libraries.

Business Intelligence Tool: Tableau Desktop / Tableau Public.

Dataset Source: Kaggle Datasets.

Presentation Tool: Microsoft PowerPoint.

Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or equivalent.

Internet Requirement: Required for downloading datasets, software installation, project hosting, dashboard publishing, and online presentations.

TECHNOLOGY

PYTHON

Python is a high-level, interpreted programming language widely used in data analytics, machine learning, artificial intelligence, and software development. It is known for its simple syntax, readability, and extensive collection of libraries. Python enables developers and analysts to perform data manipulation, statistical analysis, automation, and visualization efficiently. It supports multiple programming paradigms, including object-oriented and functional programming. Python provides powerful libraries such as Pandas, NumPy, Matplotlib, and Seaborn, which simplify data analysis and visualization tasks. Due to its flexibility, scalability, and large community support, Python has become one of the most popular programming languages in the field of Data Analytics and Data Science.

JUPYTER NOTEBOOK

Jupyter Notebook is an open-source web-based interactive computing environment used for data analysis, scientific computing, and machine learning projects. It allows users to create and share documents that contain live code, visualizations, equations, and explanatory text in a single environment. Jupyter Notebook supports multiple programming languages, with Python being the most widely used. It enables analysts to perform data cleaning, transformation, exploration, and visualization efficiently. The interactive nature of Jupyter Notebook makes it an ideal platform for developing and testing data analytics workflows while documenting the entire analytical process.

TABLEAU

Tableau is a leading business intelligence and data visualization tool used for transforming raw data into interactive and meaningful visual reports. It enables users to connect to various data sources, create charts, graphs, maps, and dashboards without extensive programming knowledge. Tableau provides drag-and-drop functionality that simplifies data exploration and visualization. It helps organizations identify trends, patterns, and business insights through interactive dashboards. Tableau supports real-time analysis, data filtering, and dashboard sharing, making it a powerful tool for decision-making and business intelligence applications.

PANDAS

Pandas is a powerful open-source Python library used for data manipulation and analysis. It provides flexible data structures such as DataFrames and Series that allow users to efficiently handle structured data. Pandas supports data cleaning, filtering, transformation, aggregation, and statistical analysis. It can read and write data from various formats including CSV, Excel, JSON, and SQL databases. Due to its efficiency and ease of use, Pandas is one of the most important libraries in the Data Analytics ecosystem.

MATPLOTLIB

Matplotlib is a comprehensive Python library used for creating static, animated, and interactive visualizations. It provides a wide variety of charts including line graphs, bar charts, histograms, scatter plots, and pie charts. Matplotlib helps analysts present data in a visually appealing and understandable manner. It integrates seamlessly with Pandas and NumPy, making it a fundamental tool for exploratory data analysis and reporting. The library offers extensive customization options for creating professional-quality visualizations.

KAGGLE DATASETS

Kaggle is an online platform that provides a vast collection of publicly available datasets for data analysis, machine learning, and research purposes. It serves as a valuable resource for data analysts and data scientists to access real-world datasets across various domains such as finance, healthcare, education, and business. Kaggle datasets enable users to practice data cleaning, analysis, visualization, and predictive modeling. The platform also promotes learning through competitions, notebooks, and collaborative projects, making it a widely used resource in the data analytics community.

IMPLEMENTATION

1. Project Planning and Requirement Analysis

Before beginning the project work, the key objectives and requirements of the Data Analytics internship were identified. The project was divided into three major components:

- **Python Project:** Development of a Stock Profit AI Optimizer to analyze stock-related data and identify potential profit opportunities.
- **Data Analytics Pipeline:** Implementation of a complete analytics workflow using Python and Jupyter Notebook, including data collection, cleaning, processing, analysis, and visualization.
- **Tableau Dashboard:** Creation of interactive dashboards and visual reports using real-world datasets obtained from Kaggle.

2. Python Project Development

The Stock Profit AI Optimizer was developed using Python programming and designed to analyze stock-related data for identifying potential profit opportunities. Various Python concepts such as variables, conditional statements, loops, functions, and object-oriented programming were utilized during development. The application processes stock data, performs analytical calculations, and presents the results in an easy-to-understand format. To provide a user-friendly interface and enable online access, the project was deployed using Streamlit, a Python-based web application framework. This deployment allowed users to interact with the application through a web browser while providing practical experience in application hosting, deployment, and project management.

3. Data Analytics Pipeline Implementation

Jupyter Notebook was used to implement the complete Data Analytics pipeline. Datasets were imported and subjected to data cleaning, preprocessing, and transformation techniques to ensure data quality. Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas and Matplotlib to identify trends, patterns, and relationships within the data. Various visualizations were created to present analytical findings in a clear and meaningful manner.

4. Tableau Dashboard Development

Tableau was used to transform raw datasets into interactive visual dashboards. A dataset obtained from Kaggle was imported into Tableau, and multiple charts, graphs, and visual reports were created. Different visualization techniques were explored to represent data effectively and highlight key insights. Finally, an interactive dashboard was developed by integrating multiple visualizations, allowing users to analyze data dynamically. The completed dashboard was presented online along with a PowerPoint presentation summarizing the project findings and outcomes.

CODING

STOCK PROFIT AI OPTIMIZER:

app.py:

```
import streamlit as st
import pandas as pd
import plotly.express as px
import kagglehub
import os
import numpy as np
from sklearn.linear_model import LinearRegression

st.set_page_config(page_title="Stock Profit AI Optimizer", page_icon="📈",
layout="wide")

@st.cache_data
def load_data():

    path =
kagglehub.dataset_download("shiivvvaam/apple-stock-market-historical-data-1980-2024")
    files = os.listdir(path)
    csv_file = [f for f in files if f.endswith('.csv')][0]
    full_path = os.path.join(path, csv_file)

    df = pd.read_csv(full_path)
    df['Date'] = pd.to_datetime(df['Date'])
    df = df.sort_values('Date')
    return df

# OPTIMIZER LOGIC (Greedy Algorithm)
def get_max_historical_profit(prices):
    """Calculates the best possible single trade in O(n) time."""
    if len(prices) < 2: return 0, 0, 0
```

```

min_price = prices[0]
max_profit = 0
buy_idx = 0
sell_idx = 0
temp_buy_idx = 0

for i, current_price in enumerate(prices):
    if current_price < min_price:
        min_price = current_price
        temp_buy_idx = i

    profit = current_price - min_price
    if profit > max_profit:
        max_profit = profit
        buy_idx = temp_buy_idx
        sell_idx = i

return max_profit, buy_idx, sell_idx

```

Linear Regression

```

def predict_future_trend(df):
    """Trains a model on recent data to predict tomorrow's price."""
    # Using the last 60 days to find a trend
    recent_data = df.tail(60).copy()
    recent_data['Day_Num'] = np.arange(len(recent_data))

    X = recent_data[['Day_Num']]
    y = recent_data['Price']

    model = LinearRegression()
    model.fit(X, y) # Model learning phase

    # Predict for the next day
    next_day_num = np.array([[len(recent_data)]])
    prediction = model.predict(next_day_num)
    return prediction[0]

```

--- 4. WEB INTERFACE (Streamlit) ---

```

def main():
    st.title("📈 Stock Profit AI Optimizer")
    st.markdown("This application uses Linear Regression to predict trends and a Greedy Algorithm to optimize historical returns.")

    try:
        df = load_data()
    except Exception as e:
        st.error(f"Failed to load data: {e}")
    return

```

```

st.sidebar.header("Settings")
year_range = st.sidebar.slider("Select Year Range",
                                int(df['Date'].dt.year.min()),
                                int(df['Date'].dt.year.max()),
                                (2015, 2024))

filtered_df = df[(df['Date'].dt.year >= year_range[0]) & (df['Date'].dt.year <=
year_range[1])].copy()

st.subheader(" Historical Optimization")
profit, b_idx, s_idx = get_max_historical_profit(filtered_df['Price'].tolist())

m1, m2, m3 = st.columns(3)
m1.metric("Max Historical Profit", f"${profit:.2f}")
m2.info(f"Best Buy Date: {filtered_df['Date'].iloc[b_idx].date()}")
m3.success(f"Best Sell Date: {filtered_df['Date'].iloc[s_idx].date()}")

st.divider()
st.subheader(" AI Trend Prediction")

pred_price = predict_future_trend(df)
current_price = df['Price'].iloc[-1]
delta = pred_price - current_price

p1, p2 = st.columns(2)
p1.metric("Predicted Price (Next Session)", f"${pred_price:.2f}", f"${delta:.2f}")

if delta > 0:
    p2.success("Model Recommendation: **BUY/LONG** (Upward Trend Detected)")
else:
    p2.warning("Model Recommendation: **HOLD/SELL** (Downward Trend Detected)")

# Visialisation
fig = px.line(filtered_df, x='Date', y='Price', title=f"Apple Stock Price
({year_range[0]}-{year_range[1]})")

fig.add_scatter(x=[filtered_df['Date'].iloc[b_idx]], y=[filtered_df['Price'].iloc[b_idx]],
                mode='markers', marker=dict(color='green', size=15, symbol='triangle-up'),
                name='Optimal Buy')
fig.add_scatter(x=[filtered_df['Date'].iloc[s_idx]], y=[filtered_df['Price'].iloc[s_idx]],
                mode='markers', marker=dict(color='red', size=15, symbol='triangle-down'),
                name='Optimal Sell')

st.plotly_chart(fig, use_container_width=True)

if st.checkbox("Show Raw Dataset"):

```

```
st.write(filtered_df.tail(100))

if __name__ == "__main__":
    main()
```

JUPYTER NOTEBOOK:

1.Importing libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2.Data Collection:

```
df_fea = pd.read_csv("features.csv")
df_train = pd.read_csv("train.csv")
df_test = pd.read_csv("test.csv")
df_store = pd.read_csv("stores.csv")
df_train_feature = pd.merge(df_train, df_fea, on=['Store', 'Date'], how='left')
df = pd.merge(df_train_feature, df_store, on='Store', how='left')
```

3.Data Understanding:

```
df.head()
df.tail()
df.shape
df.columns
df.index
df.dtypes
df.info()
df.describe()
df.describe(include='all')
df['Store'].unique()
df['Store'].nunique()
df['IsHoliday_x'].value_counts()
```

```

df.isnull().sum()
df.duplicated().sum() #np.int64(6)
df.corr(numeric_only=True)
df.memory_usage()
df.select_dtypes(include='object')
df.iloc[:,0]
df.iloc[0]
df.count()

```

4.Data Cleaning and Preprocessing :

```

df.fillna(df.mean(numeric_only=True),inplace=True)
df.duplicated().sum() #np.int64(0)

```

5.Exploratory Data Analysis (EDA) :

```

df.head()
df.info()
df.describe()
df['IsHoliday_x'].value_counts()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].mean()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].median()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].max()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].min()
df.groupby('IsHoliday_x')[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].mean()
df.groupby('IsHoliday_y')[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].mean()
df.groupby('Type')[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].mean()
df['Type'].value_counts(normalize=True)
df['Type'].unique()
df['Type'].mode()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].var()
df[['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].std()
df.groupby(['IsHoliday_x','IsHoliday_y',

```

```

'Type']][['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].mean()
df.groupby(['IsHoliday_x','IsHoliday_y',
'Type']][['MarkDown1','MarkDown2','MarkDown3','MarkDown4','MarkDown5']].agg(['mean',
'max','min'])

df.groupby(['IsHoliday_x','IsHoliday_y',
'Type'])['MarkDown1'].mean().sort_values(ascending=True)

df.groupby(['IsHoliday_x','IsHoliday_y',
'Type'])['MarkDown1'].mean().sort_values(ascending=False)

```

6.Data Visualization :

#Bar plot

```

df['Type'].value_counts().plot(kind='bar')
plt.title("Count of Store Types")
plt.show()

```

#Pie chart

```

df['IsHoliday_x'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Holiday Percentage")
plt.ylabel("")
plt.show()

```

#Histogram

```

df['MarkDown1'].plot(kind='hist')
plt.title("Distribution of MarkDown1")
plt.show()

```

#Box plot

```

df.boxplot(column='MarkDown1')
plt.title("Boxplot of MarkDown1")
plt.show()

```

#Line plot

```

df.groupby('Type')['MarkDown1'].mean().plot(kind='line')
plt.title("MarkDown1 Trend Over Time")
plt.xlabel("Type")
plt.ylabel("Average MarkDown1")
plt.show()

```

#Scatter plot

```
df.plot(kind='scatter', x='MarkDown1', y='MarkDown2')
plt.title("Scatter Plot: MarkDown1 vs MarkDown2")
plt.xlabel("MarkDown1")
plt.ylabel("MarkDown2")
plt.show()
```

#Heatmap

```
sns.heatmap(df[['MarkDown1', 'MarkDown2', 'MarkDown3', 'MarkDown4',
'MarkDown5']].corr(), annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.show()
```

#Count plot

```
sns.countplot(x='Type', data=df)
plt.title("Number of Rows per Store Type")
plt.xlabel("Store Type")
plt.ylabel("Count")
plt.show()
```

#Pair plot

```
columns = df[['MarkDown1', 'MarkDown2', 'MarkDown3']]
sns.pairplot(columns)
plt.show()
```

#Violin plot

```
filtered_df = df[df['MarkDown1'] < 20000].dropna(subset=['MarkDown1'])
sns.violinplot(x='Type', y='MarkDown1', data=filtered_df)
plt.xlabel("Store Type")
plt.ylabel("MarkDown1 Value")
plt.show()
```

#Stacked Area Chart

```
markdown_cols = ['MarkDown1', 'MarkDown2', 'MarkDown3', 'MarkDown4', 'MarkDown5']
weekly_markdowns = df.groupby('Date')[markdown_cols].mean()
weekly_markdowns.plot(kind='area', figsize=(12, 6), alpha=0.7)
plt.legend(title="Markdown Type", loc='upper left')
plt.show()
```

7.Reporting :

Observations:

- **Count plot** : Store Type A is the most frequent category in the dataset, accounting for the highest volume of records with over 200,000 rows.
- **Pair plot** : All three markdown variables (MarkDown1, MarkDown2, and MarkDown3) are heavily right-skewed. The diagonal histograms show that the vast majority of values cluster very close to zero, with a long, sparse tail extending toward high promotional amounts.
- **Heat map** : There is a strong, positive correlation of 0.79 between MarkDown1 and MarkDown4, indicating that these two promotional strategies are frequently deployed together or fluctuate similarly.

Insights:

- Insight 1: Promotional Bundling Strategy

The strong 0.79 correlation between MarkDown1 and MarkDown4 indicates that these two promotional strategies are deployed in tandem.

The company A can use these events to draw more customers.

We can also predict and optimize one performance with another.'

- Insight 2: Resource Allocation Disparity

Type A stores dominate the dataset.

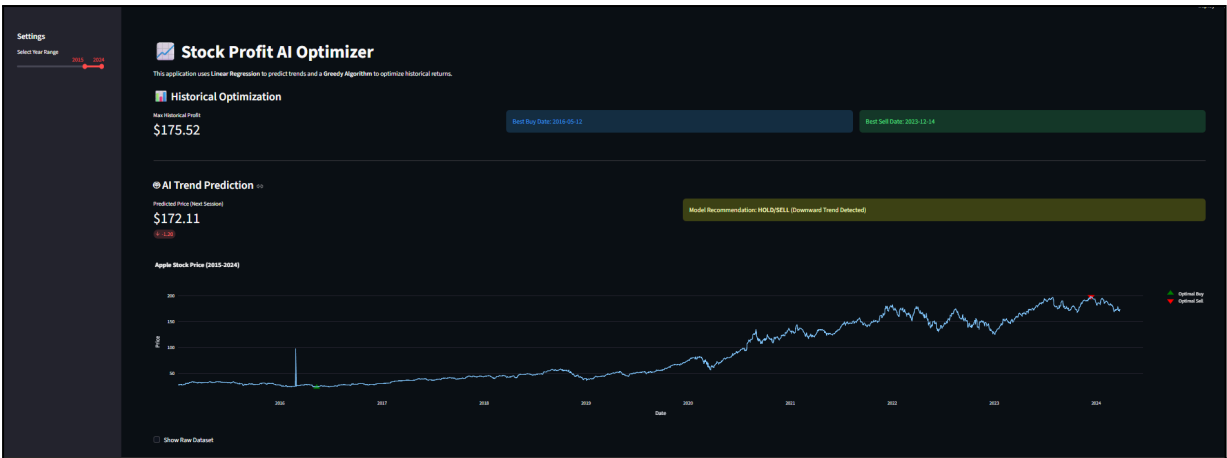
Company A receives the highest average promotional spending.

Business Recommendation:

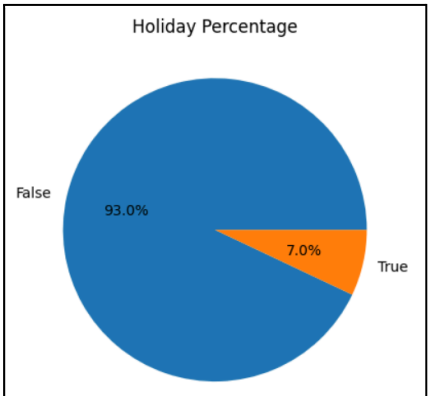
Shift 5% to 10% of the markdown budget away from Type A stores and reallocate it directly to Type C stores to capture untapped profit margins.

SAMPLE SCREENSHOTS

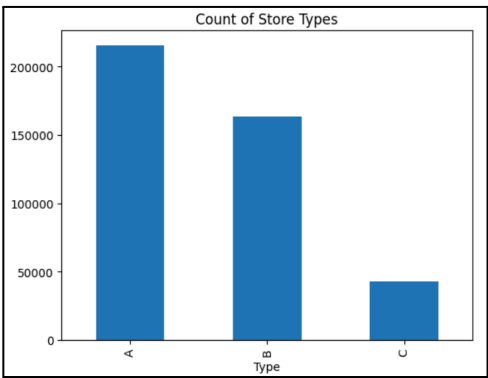
Stock Profit AI Optimizer



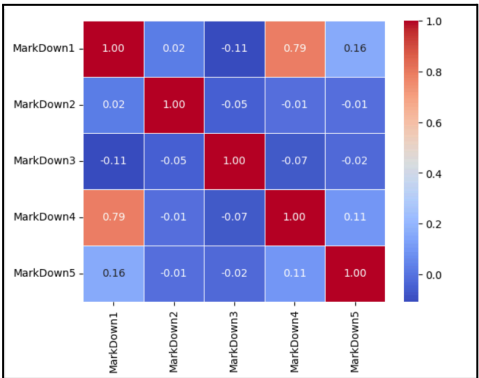
Visualization Graphs



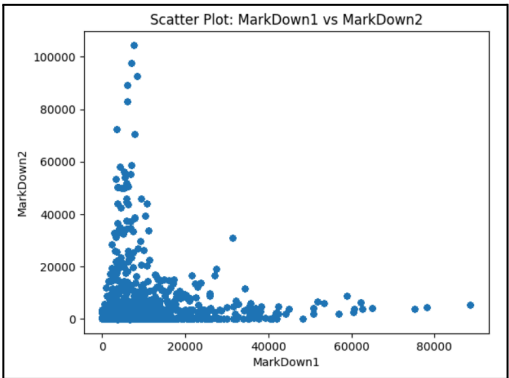
Pie Chart



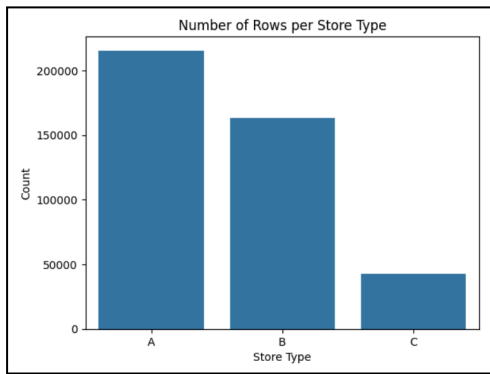
Count Plot



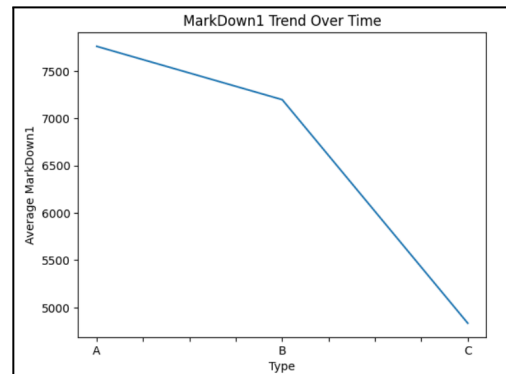
Heat Map



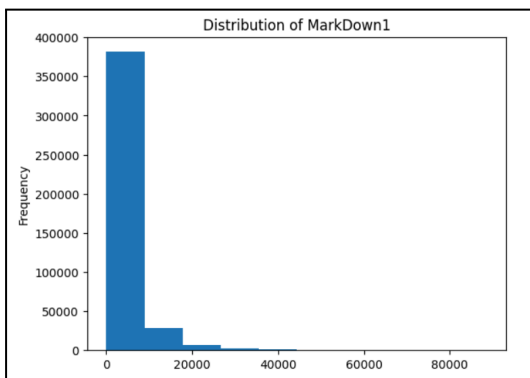
Scatter Plot



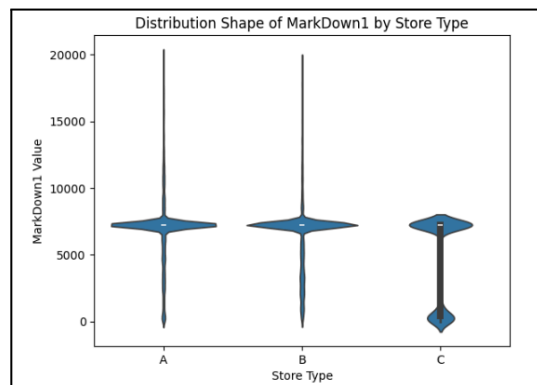
Bar Plot



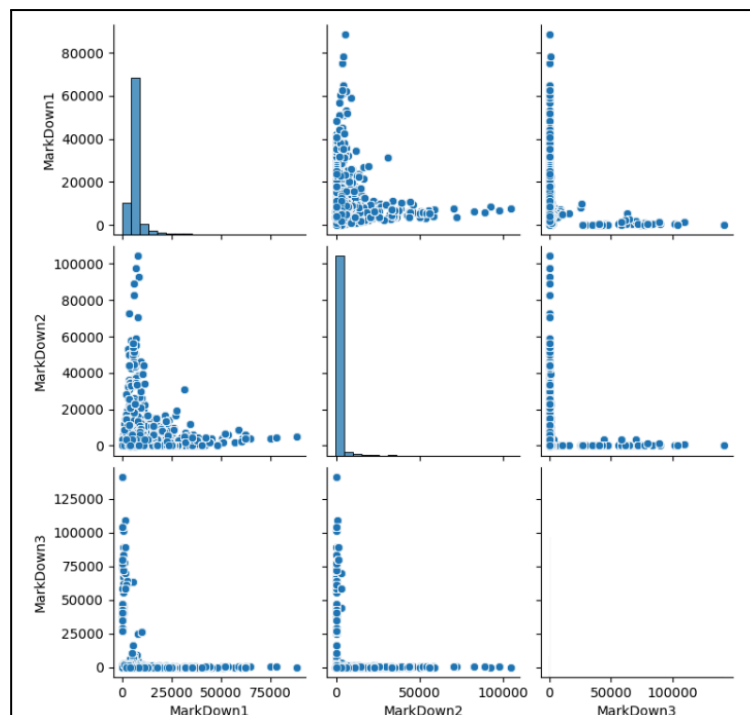
Line Plot



Histogram

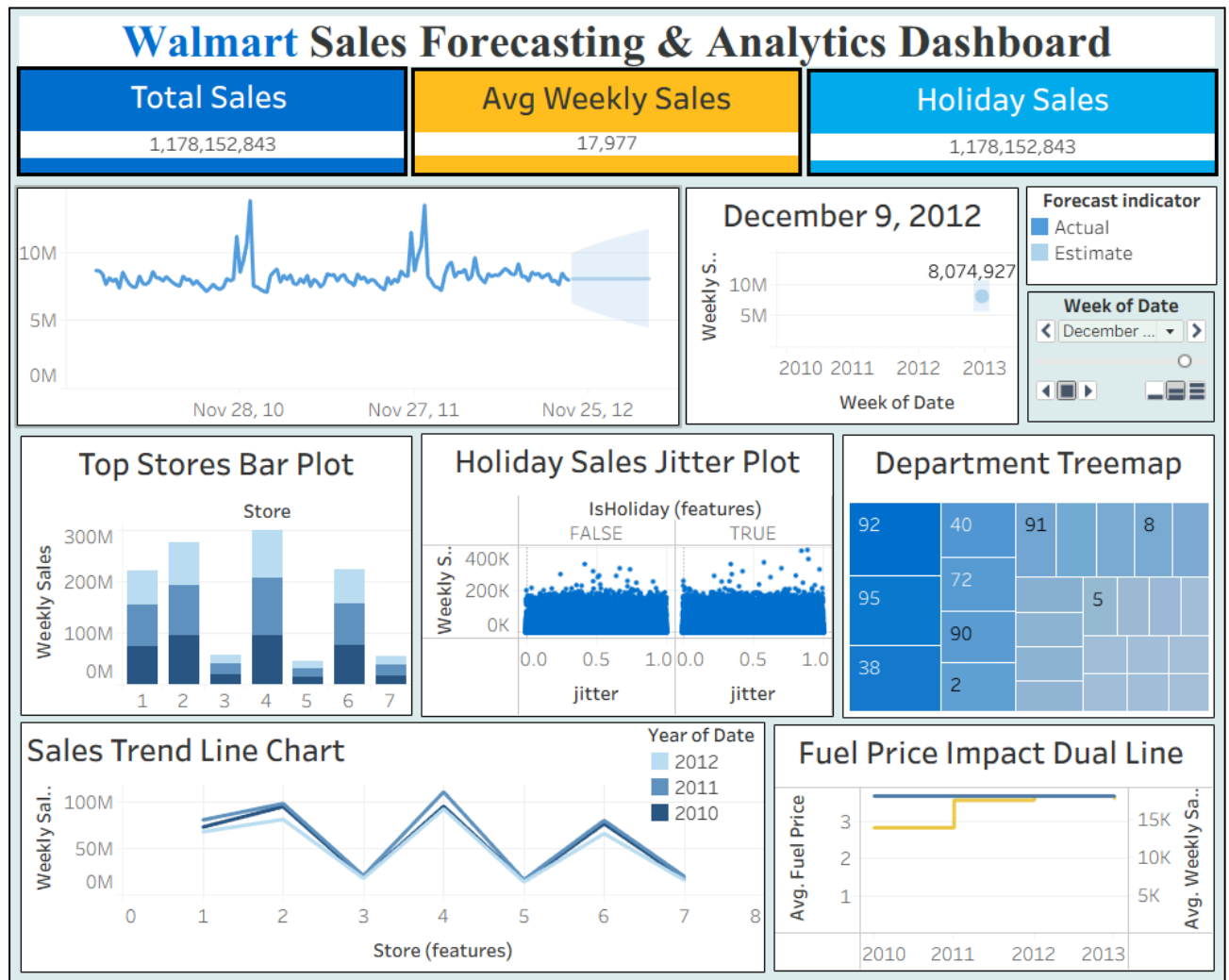


Violin Plot



Pair Plot

Dynamic Tableau Dashboard



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